

Project2Submission

Nikita's Notes (outside of results)

- Data was cleaned to remove ::: nonincremental
- Walking only has data for sessions 1 & 3
- Standing does not have data for session 2 on day 2
- Explain principle of marginality in statistical methods?
- How to justify use of LMM - discuss more...
- Add coefficients for each equation, and discuss whether used ML or REML (and impacts?)
- Explain ICC
- When discussing ICCs, just create tables to show it?? Each row is a model, and each column is a variable?
- Add final summary table to show coefficients, etc of final model; add discussion!
- discuss adding activity to ar1 - better correlation?

Section 5: Results

Section 5.1: Initial Model

Figure 1: Diagnostic Comparison of Raw & Log Linear Models

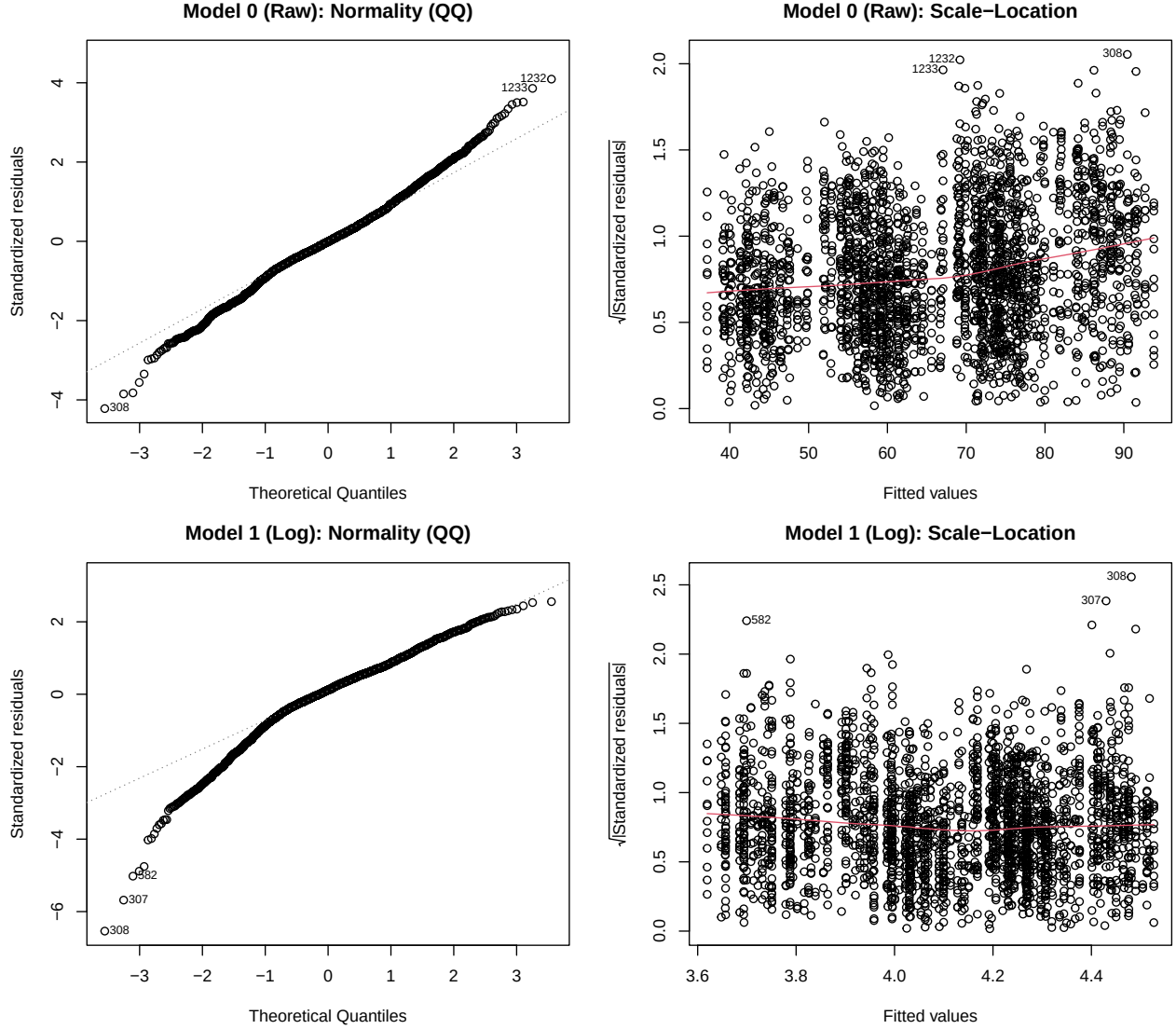


Figure 1: The QQ-plot of the Model 0 (the raw (untransformed) model) shows an approximately normal model with slightly heavy tails, but the Scale-Location plot shows heteroscedastic residuals, with variance increasing as fitted values increase. Model 1 (the log-transformed model) corrects the right heavy tail but the left tail deviates more, and the Scale-Location plot shows fairly homoscedastic residuals.

As seen in Figures 1 and 2, applying a simple linear model (Model 0: $\text{HF_value} \sim \text{session} + \text{activity} + \text{day} + \text{HF_type} + \text{sex}$) shows that the residuals are approximately normal (with heavy tails), but heteroscedastic. Transforming the response variable (Model 1: $\log(\text{HF_value})$) resulted in homoscedasticity and corrected the heavy right tail, but introduced more left-skew in the residuals. We prioritize homoscedasticity and use Model 1 (the log model) going forward, to ensure reliable parameter estimation and significance testing.

Table 1: Mathematical Diagnostic Comparison of Model 0 & Model 1

Metric	Model 0 (Raw)	Model 1 (Log)
Normality	0.9966	0.9768
Heteroscedasticity	0.0274	-0.0524

Table 2: Backwards Selection of Fixed Effects: Likelihood Ratio Test Summary

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 1 (Full Model)</i>	14	1499	1581	-736	–	–
Drop Day	13	1755	1832	-865	258.01	< 0.0001
Drop Session	11	1498	1563	-738	5.04	0.169116376183573
Drop Activity	12	1506	1576	-741	10.78	0.00456676439709016
Drop Sensor	9	2438	2491	-1210	948.97	< 0.0001
Drop Sex	13	1506	1582	-740	8.77	0.00305399944413452

Table 2: Model selection of fixed effects via backwards elimination. Model 1 (the 'Full Model') includes all predictors. Each subsequent row represents a model where only one term was dropped from Model 1. The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced model to Model 1. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Table 1: Mathematical diagnostics comparison between Model 0 (raw) and Model 1 (log-transformed model), comparing normality (correlation between actual and theoretically normal residuals, with values closer to 1 indicating normality) and heteroscedasticity (correlation between fitted values and residuals, with values closer to 0 indicating heteroscedasticity). While Model 1 shows slightly worse normality, it is significantly closer to homoscedasticity.

Section 5.2: Fixed Effects

As seen in Table 2, although Model 1 includes all fixed effects ($\log(\text{HF_value}) \sim \text{session} + \text{activity} + \text{day} + \text{sensor} + \text{sex}$), a backwards selection process was conducted to ensure parsimony. We evaluated each variable by comparing the full model to nested models where one term was removed, using the Likelihood Ratio Test (LRT) and changes in information criteria (AIC and BIC).

At first glance, all variables are statistically significant except for the model dropping the session variable ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$), as the p-value from the likelihood ratio test of this model versus the full model is not below 0.05 ($p=0.1691$). Additionally, although the models that drop activity and drop sex are technically statistically significant ($p\text{-values} < 0.01$), these models show a unique characteristic of having slightly higher AICs than the full model, but lower or similar BICs. This discrepancy implies that although these variables improve prediction, the improvement may not be large enough to justify including the variables. In this case (and similar cases further in the analysis), the variables are included but will be re-examined in final model pruning.

As such, the final fixed effect model was Model 2 ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$).

Section 5.3: Interaction Terms

We further examined fixed effects by examining the value of adding all possible interaction terms to the model, as shown in Table 3. Testing adding each interaction term against against Model 2 ($\text{HF_value} \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$), we saw that adding the following interaction terms results in models with worse (or similar) AIC & BIC than the current model without interaction terms, and thus did not add them to the model: Activity & Day, Activity & Sex, Session & Sex, Day & Sex, Day & Session.

Technically, adding the interaction of Day & Session results in a p-value of 0.0118, below 0.05. However, this interaction increased the BIC and barely decreases the AIC, so we chose not to include it in the model. Note that the interaction of Activity & Session was not tested - because there is no data for certain interactions (ex. walking & session 2, walking & session 4), the model cannot run.

The interaction of Sensor with the other possible variables (Day, Activity, Sex, and Session) all have low p-values (<0.001) and result in decreased AIC, although two interaction terms (Sensor and Sex, and Sensor and Session) result in increased BIC. Table 2 showed a similar story - both Session and Sex decreased AIC slightly, but increased BIC. Similar to Section 5.1, we include these interaction terms for now and will prune the model at the end.

Table 3: Interaction Terms: Likelihood Ratio Test Summary

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 2 (No Interaction Terms)</i>	11	1498	1563	-738	–	–
Interact Sensor and Day	16	1311	1405	-640	196.89	< 0.0001
Interact Sensor and Activity	21	1340	1463	-649	178.67	< 0.0001
Interact Sensor and Sex	16	1482	1576	-725	25.78	< 0.0001
Interact Sensor and Session	29	1455	1625	-698	79.58	< 0.0001
Interact Activity and Day	13	1502	1578	-738	0.38	0.828585217668789
Interact Activity and Sex	13	1499	1575	-736	3.43	0.17993819517717
Interact Session and Sex	17	1504	1604	-735	6.53	0.366542390591828
Interact Day and Sex	12	1500	1570	-738	0.63	0.426622022410979
Interact Day and Session	17	1494	1594	-730	16.4	0.0117744366721763

Table 3: The top row (Model 2) is the row without any interaction terms. Each subsequent row represents a model where only one interaction term was added to Model 2. The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the new model to Model 2 via ANOVA. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Table 4: Backwards Selection of Interaction Terms: Likelihood Ratio Test Summary

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 3 (Four Interaction Terms)</i>	49	1107	1395	-505	NA	NA
Drop Sensor and Day	44	1314	1573	-613	216.74	0.0000000
Drop Sensor and Activity	39	1248	1477	-585	160.62	0.0000000
Drop Sensor and Sex	44	1127	1386	-520	30.02	0.0000146
Drop Sensor and Session	31	1113	1295	-525	41.49	0.0013039

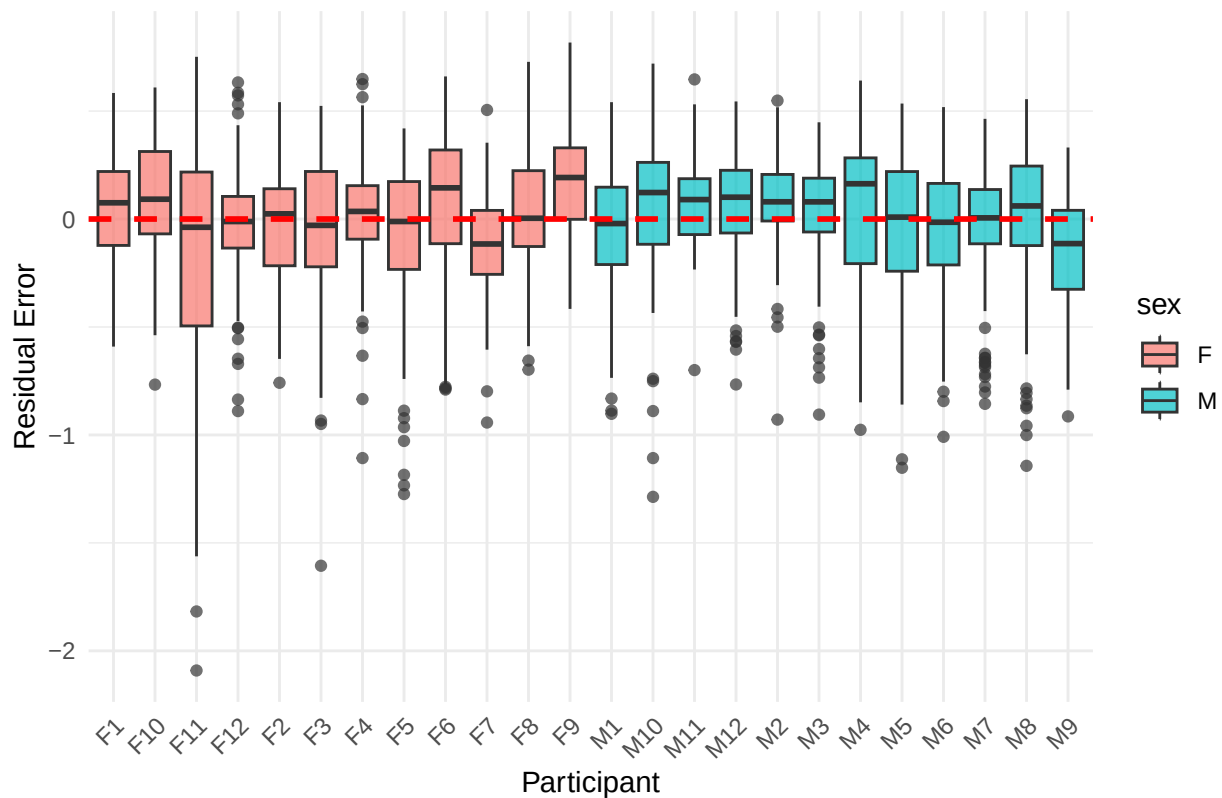
Table 4: Model selection of interaction terms via backwards elimination. Model 3 includes the four interaction terms previously noted. Each subsequent row represents a model where only one interaction term was dropped from Model 3. Note that for the last row, Drop Sensor and Session, the fixed effect Session is dropped as well (Session was only included in Model 3 because the interaction of Sensor and Session was added). The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced model to Model 3. Note all models were fit using Maximum Likelihood (ML), to compare across models.

This results in Model 3: $\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex} + \text{sensor} * \text{activity} + \text{sensor} * \text{day} + \text{sensor} * \text{sex}$. Note that session is added back into the model automatically, as the interaction of sensor and session is included, due to the principle of marginality. Table 4 shows the results of checking if removing any of the four interaction terms improves Model 3. This showed a similar story to the individual tests of adding interaction terms - the inclusion of the interactions of Sensor and Sex, and Sensor and Session, slightly decrease AIC but increase BIC.

Section 5.4: Random Intercepts

TO DO: ADD JUSTIFICATION FOR LMM

Residuals by Person for Model 1, fixed-effect-only model



Motivation for LMM: Residuals by Person (Model 3)

Systematic shifts away from zero prove that errors within an individual are correlated.

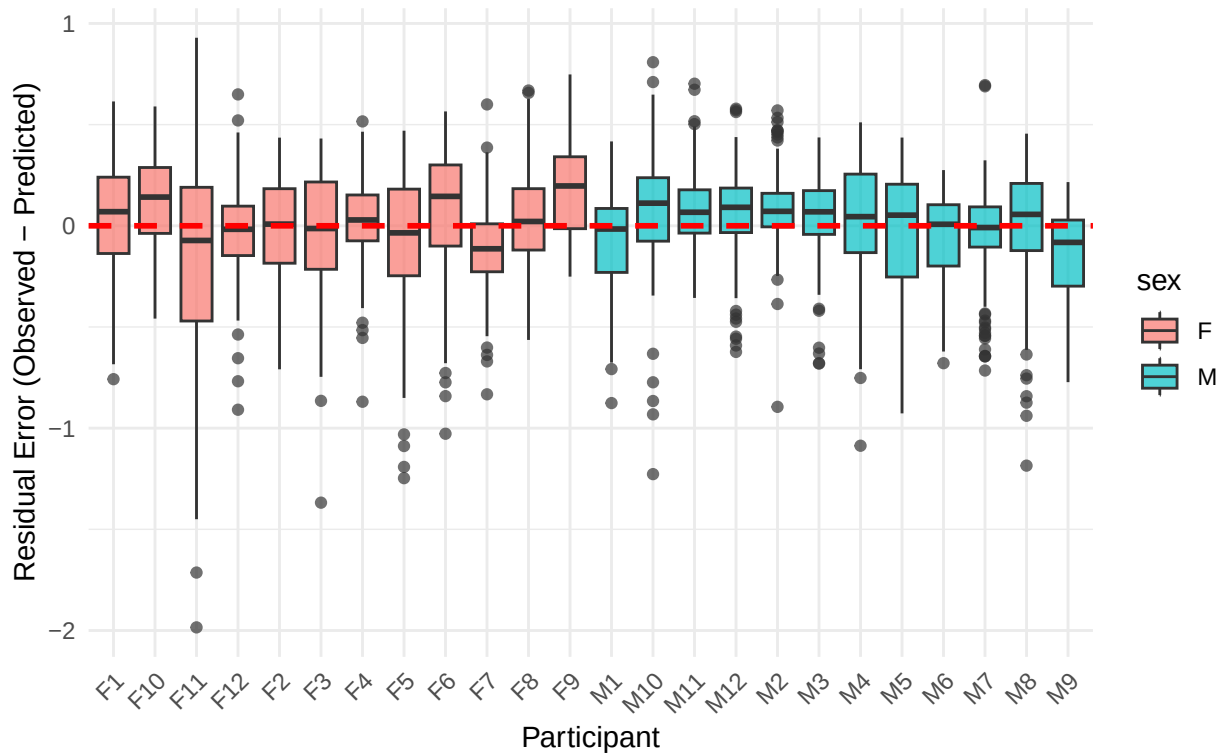


Table 5: Likelihood Ratio Test for Random Intercept for Person

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 3	49	1356	1643	-629	NA	NA
Model 4	50	1208	1501	-554	149.96	0

Table 5: Impact of adding a random intercept for person (Model 4) to the fixed-effect only model (Model 3). Note this uses REML to compare models, as fixed effects are not changing. As such, the final p-values should be divided by 2. Given the p-value is already extremely low, this has no impact on the takeaway that we should include the Random Intercept for Person. Note models were fit using Restricted Maximum Likelihood (REML), as fixed effects are the same between models.

To determine if there is inter-individual variability, we added a random intercept for each person, creating Model 4: $\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex} + \text{sensor} * \text{activity} + \text{sensor} * \text{day} + \text{sensor} * \text{sex} + \text{sensor} * \text{session} +$. This assumes that all individuals have unique baselines for heat flow, regardless of if they are performing the same activity and/or are at the same temperature. Note the models in this section (5.4) were run using REML to ensure estimates of variance components are not biased, unless explicitly stated otherwise.

The Intraclass Correlation Coefficient (ICC) for Model 4 was 0.0791, implying that ~8% of the variance is due to differences between people (and 92% is unexplained). This suggests there is some inter-individual variability. Table 5 shows that this also reduces AIC and BIC significantly,

We then examined if a nested activity-level random effect was necessary. The ICC for person remains the same as in model 4 (0.0791), and the ICC of activity (within person) is 0, implying adding activity as a random intercept adds no value - the effect of activity does not vary randomly between people.

We then examined if a nested sensor-level random intercept was necessary (Model 5). The ICC for person is now 0.0184, lower than in model 4 (0.0791), and the ICC of sensor (within person) is 0.3772, with 60% of the variance unexplained. This implies that the major source of clustering is at the sensor level, and not individual level.

Removing the person random intercept (resulting in a sensor-only random intercept model) results in the significantly lower ICC for sensor (0.0159), with 98% of the variance left unexplained. This implies that the sensor by itself is not the primary driver of variance; instead, there is a shared baseline amongst sensors belonging to the same individual. Thus, we keep the nested sensor-level random intercept model (model 5).

Section 5.5: Random Slopes

The current model has random intercepts for person and sensor (nested), implying that measurements from the same sensor are correlated, and sensors belonging to the same person are correlated. However, it assumes that the impact of temperature (the day variable) is identical for all individuals. We tested day as a random slope (given person) to determine if the effect of day varies by person, meaning some people adapt to the temperature environment differently than others.

Adding day as a random slope (Model 6) results in 45% unexplained variance, compared to 60% from Model 5. The ICC for day is 0.186, which is fairly large; ~19% of the variation in heat flow measurements is explained by the fact that people have different heat flow reactions depending on the external temperature. Additionally, the correlation between the person-level intercept and the day-level random slope is -0.884. This implies that the higher a person's heat flow was on day 1 (hot day), the steeper their drop was on day 2 (cold day), suggesting convergence.

The ICC for person also increased to 0.0419 (compared to 0.0184 from model 5), suggesting that the noise from people reacting differently to temperature masked their baseline differences. The ICC for sensor decreased to 0.3241 (from 0.3772 in model 5), suggesting that temperature explained some of the heat flow variance that was previously attributed to sensor.

Table 6: Likelihood Ratio Test for Random Slope for day

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 5: Random Intercept: Person and Nested Sensor	51	485	784	-191	–	0
Model 6: Add Random Slope for Day (Given Person)	53	312	622	-103	176.82	0

Table 6: Likelihood test between Model 5 and Model 6, which adds day as a random slope. Model 6 has significantly lower AIC and BIC, and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects..

Table 7: Likelihood Ratio Test for Covariance Structures

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 6: Assumes Compound Symmetry	53	312	622	-103	–	0
Model 7: Assumes Auto-regressive AR(1)	54	-402	-86	255	715.88	0

Table 7: Likelihood test between Model 6 and Model 7, testing an AR1 structure where magnitude of correlation depends on amount of time between observations (instead of flat correlation). Model 7 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Section 5.7: Covariance Structures

The selected hierarchical structure (Model 6) effectively partitions the data’s variance into three distinct components: global participant-level baselines, individual thermal adaptation patterns (represented by the random slope for day), and nested sensor-level deviations. This framework accounts for the inherent dependencies in the data while acknowledging that heat flow sensitivity is not uniform across the study population.

However, this still imposes a strict assumption of conditional independence for the remaining residual errors. It has a compound symmetry structure, assuming a constant correlation structure where the correlation between Session 1 and Session 2 is identical to the correlation between Session 1 and Session 4, ignoring the fact that observations closer in time are typically more highly correlated. As such, we examined implementing an autoregressive (AR1) correlation structure for the residuals in Model 7. We assumed this was nested within person, sensor, and day, ensuring independence between people, sensor location, and days. We did not assume independence between activity because the activities were done in order, with session 1 consisting of sitting, then standing, then walking, then a break, and then session two beginning with the same cycle (although sessions 2 and 4 did not have walking). This assumes that these measurements were equally spaced in time, which is likely not true - timestamp data could improve this assumption.

The correlation structure has a ϕ of 0.5745, meaning that the correlation between session 1 and 2 is ~ 0.57 , and session 1 & 3 is $\sim 0.57^2$. This is fairly high, suggesting the previous data point strongly influences the next one. Since we have the data segmented by session and activity, it suggests that sitting in session 1 impacts standing in session 1, and walking in session 1 impacts sitting in session 2. Additionally, Table 7 shows that Model 7 has significantly lower AIC and BIC, and is statistically significant in the likelihood ratio test against Model 6, suggesting an AR1 covariance structure is a significant improvement on the Compound Symmetry structure, and that there is a strong temporal auto-correlation.

Model 7 (AR(1)) showed that there is a temporal aspect to the data, where consecutive sessions are more closely linked than further ones. However, AR(1) assumes that the amount of variance is identical for all sensors. However, we would expect sensors on different body parts to have different variance. As such, we

Table 7: Likelihood Ratio Test for Covariance Structures

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 6: Assumes Compound Symmetry	53	312	622	-103	–	0
Model 7: Assumes Auto-regressive AR(1)	54	-402	-86	255	715.88	0

Table 7: Likelihood test between Model 6 and Model 7, testing an AR1 structure where magnitude of correlation depends on amount of time between observations (instead of flat correlation). Model 7 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Table 7: Likelihood Ratio Test for Covariance Structures

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 6: Assumes Compound Symmetry	53	312	622	-103	—	0
Model 7: Assumes Auto-regressive AR(1)	54	-402	-86	255	715.88	0

Table 8: Likelihood test between Model 7 and Model 8, testing an ARH1 structure that allows each sensor to have its own unique residual variance instead of assuming uniform residual noise. Model 8 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Table 9: Backwards Selection of Interaction Terms and Variables: Likelihood Ratio Test Summary

Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 8 (Full Model)</i>	<i>59</i>	<i>-1193</i>	<i>-846</i>	<i>655</i>	<i>NA</i>	<i>NA</i>
Drop Sensor and Sex Interaction	54	-1198	-881	653	4.13	0.5305701
Drop Sex Fully	53	-1199	-887	652	5.56	0.4737540
Drop Sensor and Day Interaction	54	-1135	-818	622	67.53	0.0000000
Drop Day Fully	53	-1107	-795	606	97.93	0.0000000
Drop Sensor and Activity Interaction	49	-809	-521	453	403.85	0.0000000
Drop Activity Fully	47	-769	-493	432	447.07	0.0000000
Drop Sensor and Session Interaction	44	-1187	-928	638	35.49	0.0020929
Drop Session Fully	41	-1190	-949	636	38.39	0.0034388

Table 9: Model selection of variables via backwards elimination. Model 8 includes the four interaction terms previously noted. Each subsequent row represents a model where either only one interaction term was dropped from Model 8 (Drop X and Y Interaction), or the whole variable and interaction (Drop Sex Fully means dropped the variable sex, and the interaction term sex and sensor). The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced models to Model 8. Note all models were fit using Maximum Likelihood (ML), to compare across models.

tested a heterogeneous AR structure ARH(1) by combining Model 7’s AR(1) structure with heterogeneous residual variances across sensors.

Model 8’s ϕ is 0.6269, higher than Model 7’s (0.5745), suggesting the temporal aspect of the data was higher than previously suggested and was masked by the noise levels of the sensors. The estimates for the heterogenous variances relative to sensor 16 (baseline, dominant hand) are: 1.27 (sensor 2, chest (side)), 0.72 (sensor 3, upper chest), 0.35 (sensor 5, upper arm), 0.95 (sensor 7, lower arm), and 0.97 (sensor 8, abdomen). This implies that sensors on the abdomen, lower arm, and dominant hand have similar noise. The sensor on the chest (side) is significantly noisier than those sensors, and the sensors on the upper chest and arm are significantly less noisy. There does not seem to be an obvious trend or reason for this; it may be helpful to consult experts in the field. Additionally, Table 8 shows that the ARH(1) structure (Model 8) has significantly lower AIC and BIC than Model 7’s AR(1) structure, and is statistically significant in the likelihood ratio test.

```
##           Model df          AIC          BIC    logLik    Test  L.Ratio p-value
## mdl7_reml      1  54 -401.9296   -85.5213  254.9648
## mdl8_reml      2  59 -943.6603  -597.9549  530.8301  1 vs 2  551.7307  <.0001
```

Section 5.8: Pruning Fixed Effects

The final model utilizes an ARH(1) covariance structure to account for the temporal aspect of the data between sequential activities and sessions, while allowing for unique levels of noise at each sensor. Additionally, it has random intercepts for individuals and nested sensors to account for sensor-specific baselines per individual, and a person-level random slope for day to capture individual level responses to temperature changes. The model has become quite complex, so we re-examine fixed effects to determine if there are any that can be removed.

This suggests that we should remove the variables Sex and the interaction term of Sensor and Sex, as the model without those terms has better AIC and BIC, and the p-value from the likelihood ratio test is not statistically significant (>0.4). We also removed the variable Session and the interaction term of Sensor and

Session. Although they minimally improve the AIC (by <10 points), they worsen the BIC significantly and introduce ~18 degrees of freedom.

This results in the final Model 9: $\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sensor} * \text{activity} + \text{sensor} * \text{day}$, which evaluates the fixed effects of activity, day, sensor, and the interactions of sensor with activity and day. It addresses the hierarchical nature of the data by incorporating a random intercept for person, a nested random intercept for sensor, and a person-level random slope for day. Finally, the error structure is defined by an ARH(1) covariance structure, allowing temporal auto-correlation segmented by person, sensor, and day, and heterogeneous residual variances to account for different levels of volatility across sensors.

```
## Linear mixed-effects model fit by REML
##   Data: data_clean
##       AIC       BIC    logLik
##   -1037.279 -796.7589 559.6394
##
## Random effects:
##   Formula: ~1 + day | person
##   Structure: General positive-definite, Log-Cholesky parametrization
##           StdDev      Corr
## (Intercept) 0.08198097 (Intr)
## day2        0.13882909 -0.922
##
##   Formula: ~1 | HF_type %in% person
##           (Intercept) Residual
## StdDev:   0.1591161 0.2849547
##
## Correlation Structure: AR(1)
##   Formula: ~1 | person/HF_type/day
##   Parameter estimate(s):
##       Phi
## 0.6324753
## Variance function:
##   Structure: Different standard deviations per stratum
##   Formula: ~1 | HF_type
##   Parameter estimates:
##       HF_16    HF_2    HF_3    HF_5    HF_7    HF_8
## 1.0000000 1.2388910 0.7058033 0.3412306 0.9310399 0.9463243
## Fixed effects: log(HF_value) ~ (day + activity + sex) * HF_type
##
##              Value Std.Error   DF   t-value p-value
## (Intercept)      4.382541 0.06525206 2476   67.16326 0.0000
## day2           -0.146868 0.05693764 2476   -2.57945 0.0100
## activitystanding -0.077875 0.01749847 2476   -4.45040 0.0000
## activitywalking   0.094058 0.02127396 2476    4.42128 0.0000
## sexM             -0.003530 0.08263706   22   -0.04272 0.9663
## HF_typeHF_2      -0.570742 0.09382998  110   -6.08273 0.0000
## HF_typeHF_3      -0.538773 0.08400422  110   -6.41365 0.0000
## HF_typeHF_5      -0.317337 0.07947313  110   -3.99301 0.0001
## HF_typeHF_7      -0.116802 0.08715457  110   -1.34018 0.1830
## HF_typeHF_8      -0.781547 0.08765932  110   -8.91573 0.0000
## day2:HF_typeHF_2   0.470108 0.07826288 2476    6.00679 0.0000
## day2:HF_typeHF_3   0.467161 0.06086916 2476    7.67484 0.0000
## day2:HF_typeHF_5   0.412815 0.05278869 2476    7.82013 0.0000
## day2:HF_typeHF_7   0.308993 0.06682584 2476    4.62386 0.0000
## day2:HF_typeHF_8   0.464246 0.06796492 2476    6.83067 0.0000
## activitystanding:HF_typeHF_2 -0.105925 0.02754265 2476   -3.84585 0.0001
```

```

## activitywalking:HF_typeHF_2 -0.461892 0.03378722 2476 -13.67060 0.0000
## activitystanding:HF_typeHF_3 0.178397 0.02139428 2476 8.33853 0.0000
## activitywalking:HF_typeHF_3 0.170678 0.02612520 2476 6.53307 0.0000
## activitystanding:HF_typeHF_5 0.051458 0.01850259 2476 2.78110 0.0055
## activitywalking:HF_typeHF_5 -0.110354 0.02252214 2476 -4.89980 0.0000
## activitystanding:HF_typeHF_7 -0.037174 0.02362961 2476 -1.57320 0.1158
## activitywalking:HF_typeHF_7 -0.144755 0.02890232 2476 -5.00842 0.0000
## activitystanding:HF_typeHF_8 0.162279 0.02399148 2476 6.76404 0.0000
## activitywalking:HF_typeHF_8 0.091377 0.02923513 2476 3.12559 0.0018
## sexM:HF_typeHF_2 -0.072653 0.12069664 110 -0.60195 0.5484
## sexM:HF_typeHF_3 0.069682 0.11020538 110 0.63229 0.5285
## sexM:HF_typeHF_5 0.019993 0.10593391 110 0.18873 0.8507
## sexM:HF_typeHF_7 0.130835 0.11358630 110 1.15186 0.2519
## sexM:HF_typeHF_8 0.093280 0.11428393 110 0.81621 0.4161
## Correlation:
## (Intr) day2 actvtys actvtyw sexM HF_HF_2
## day2 -0.433
## activitystanding -0.134 0.013
## activitywalking -0.097 -0.001 0.408
## sexM -0.618 -0.019 -0.001 0.001
## HF_typeHF_2 -0.643 0.219 0.093 0.067 0.419
## HF_typeHF_3 -0.718 0.245 0.104 0.075 0.468 0.500
## HF_typeHF_5 -0.759 0.259 0.110 0.079 0.494 0.528
## HF_typeHF_7 -0.692 0.236 0.101 0.072 0.451 0.481
## HF_typeHF_8 -0.688 0.235 0.100 0.072 0.448 0.479
## day2:HF_typeHF_2 0.229 -0.547 -0.009 0.001 0.014 -0.402
## day2:HF_typeHF_3 0.295 -0.704 -0.012 0.001 0.018 -0.206
## day2:HF_typeHF_5 0.340 -0.812 -0.014 0.001 0.022 -0.238
## day2:HF_typeHF_7 0.269 -0.641 -0.011 0.001 0.017 -0.187
## day2:HF_typeHF_8 0.264 -0.630 -0.011 0.001 0.017 -0.185
## activitystanding:HF_typeHF_2 0.085 -0.008 -0.635 -0.259 0.001 -0.147
## activitywalking:HF_typeHF_2 0.061 0.000 -0.257 -0.630 -0.001 -0.105
## activitystanding:HF_typeHF_3 0.110 -0.010 -0.818 -0.334 0.001 -0.076
## activitywalking:HF_typeHF_3 0.079 0.001 -0.332 -0.814 -0.001 -0.055
## activitystanding:HF_typeHF_5 0.127 -0.012 -0.946 -0.386 0.001 -0.088
## activitywalking:HF_typeHF_5 0.091 0.001 -0.385 -0.945 -0.001 -0.063
## activitystanding:HF_typeHF_7 0.099 -0.009 -0.741 -0.302 0.001 -0.069
## activitywalking:HF_typeHF_7 0.071 0.001 -0.300 -0.736 -0.001 -0.049
## activitystanding:HF_typeHF_8 0.098 -0.009 -0.729 -0.297 0.001 -0.068
## activitywalking:HF_typeHF_8 0.070 0.001 -0.297 -0.728 -0.001 -0.049
## sexM:HF_typeHF_2 0.412 0.013 0.001 -0.001 -0.668 -0.625
## sexM:HF_typeHF_3 0.452 0.015 0.001 -0.001 -0.731 -0.314
## sexM:HF_typeHF_5 0.470 0.015 0.001 -0.001 -0.761 -0.326
## sexM:HF_typeHF_7 0.438 0.014 0.001 -0.001 -0.709 -0.305
## sexM:HF_typeHF_8 0.435 0.014 0.001 -0.001 -0.705 -0.302
## HF_HF_3 HF_HF_5 HF_HF_7 HF_HF_8 d2:HF_HF_2
## day2
## activitystanding
## activitywalking
## sexM
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5 0.590
## HF_typeHF_7 0.538 0.568

```

```

## HF_typeHF_8          0.535    0.565    0.515
## day2:HF_typeHF_2     -0.179   -0.189   -0.172   -0.172
## day2:HF_typeHF_3     -0.358   -0.244   -0.221   -0.221    0.514
## day2:HF_typeHF_5     -0.265   -0.325   -0.255   -0.255    0.594
## day2:HF_typeHF_7     -0.209   -0.221   -0.368   -0.200    0.466
## day2:HF_typeHF_8     -0.206   -0.218   -0.198   -0.374    0.461
## activitystanding:HF_typeHF_2 -0.066   -0.070   -0.064   -0.063    0.014
## activitywalking:HF_typeHF_2 -0.047   -0.050   -0.046   -0.045    0.000
## activitystanding:HF_typeHF_3 -0.128   -0.090   -0.082   -0.082    0.007
## activitywalking:HF_typeHF_3 -0.091   -0.065   -0.059   -0.059    0.000
## activitystanding:HF_typeHF_5 -0.099   -0.116   -0.095   -0.094    0.009
## activitywalking:HF_typeHF_5 -0.071   -0.084   -0.068   -0.068   -0.001
## activitystanding:HF_typeHF_7 -0.077   -0.082   -0.136   -0.074    0.007
## activitywalking:HF_typeHF_7 -0.055   -0.058   -0.097   -0.053    0.000
## activitystanding:HF_typeHF_8 -0.076   -0.080   -0.073   -0.137    0.007
## activitywalking:HF_typeHF_8 -0.055   -0.058   -0.053   -0.098    0.000
## sexM:HF_typeHF_2     -0.320   -0.338   -0.309   -0.306   -0.022
## sexM:HF_typeHF_3     -0.646   -0.370   -0.338   -0.336   -0.012
## sexM:HF_typeHF_5     -0.364   -0.653   -0.352   -0.349   -0.013
## sexM:HF_typeHF_7     -0.340   -0.360   -0.644   -0.326   -0.010
## sexM:HF_typeHF_8     -0.338   -0.357   -0.326   -0.636   -0.011
##                      d2:HF_HF_3 d2:HF_HF_5 d2:HF_HF_7 d2:HF_HF_8
## day2
## activitystanding
## activitywalking
## sexM
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5      0.763
## day2:HF_typeHF_7      0.600      0.692
## day2:HF_typeHF_8      0.592      0.684      0.537
## activitystanding:HF_typeHF_2 0.007      0.009      0.007      0.007
## activitywalking:HF_typeHF_2 0.000     -0.001      0.000      0.000
## activitystanding:HF_typeHF_3 0.015      0.011      0.009      0.009
## activitywalking:HF_typeHF_3 -0.001     -0.001     -0.001     -0.001
## activitystanding:HF_typeHF_5 0.011      0.014      0.010      0.010
## activitywalking:HF_typeHF_5 -0.001     -0.001     -0.001     -0.001
## activitystanding:HF_typeHF_7 0.009      0.010      0.014      0.008
## activitywalking:HF_typeHF_7 0.000     -0.001      0.000      0.000
## activitystanding:HF_typeHF_8 0.009      0.010      0.008      0.014
## activitywalking:HF_typeHF_8 0.000     -0.001      0.000      0.000
## sexM:HF_typeHF_2     -0.014   -0.017   -0.011   -0.013
## sexM:HF_typeHF_3     -0.014   -0.019   -0.013   -0.014
## sexM:HF_typeHF_5     -0.016   -0.022   -0.013   -0.015
## sexM:HF_typeHF_7     -0.013   -0.015   -0.013   -0.012
## sexM:HF_typeHF_8     -0.015   -0.018   -0.012   -0.022
##                      actvtys:HF_HF_2 actvtys:HF_HF_2 actvtys:HF_HF_3
## day2
## activitystanding

```

```

## activitywalking
## sexM
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2 0.405
## activitystanding:HF_typeHF_3 0.520 0.210
## activitywalking:HF_typeHF_3 0.211 0.513 0.407
## activitystanding:HF_typeHF_5 0.601 0.243 0.774
## activitywalking:HF_typeHF_5 0.245 0.595 0.315
## activitystanding:HF_typeHF_7 0.470 0.190 0.606
## activitywalking:HF_typeHF_7 0.191 0.463 0.245
## activitystanding:HF_typeHF_8 0.463 0.187 0.597
## activitywalking:HF_typeHF_8 0.189 0.458 0.243
## sexM:HF_typeHF_2 0.000 0.001 0.000
## sexM:HF_typeHF_3 0.000 0.001 0.000
## sexM:HF_typeHF_5 0.000 0.001 -0.001
## sexM:HF_typeHF_7 0.000 0.001 -0.001
## sexM:HF_typeHF_8 0.000 0.001 0.000
## actvttyw:HF_HF_3 actvtty:HF_HF_5 actvttyw:HF_HF_5
## day2
## activitystanding
## activitywalking
## sexM
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5 0.314
## activitywalking:HF_typeHF_5 0.769 0.407
## activitystanding:HF_typeHF_7 0.246 0.700 0.285
## activitywalking:HF_typeHF_7 0.599 0.284 0.695
## activitystanding:HF_typeHF_8 0.242 0.690 0.281
## activitywalking:HF_typeHF_8 0.593 0.281 0.687
## sexM:HF_typeHF_2 0.001 -0.001 0.001
## sexM:HF_typeHF_3 0.000 -0.001 0.001

```

```

## sexM:HF_typeHF_5          0.001          -0.001          0.001
## sexM:HF_typeHF_7          0.001          -0.001          0.001
## sexM:HF_typeHF_8          0.001          -0.001          0.001
##                          actvtys:HF_HF_7 actvttyw:HF_HF_7 actvtys:HF_HF_8
## day2
## activitystanding
## activitywalking
## sexM
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7  0.408
## activitystanding:HF_typeHF_8  0.540          0.219
## activitywalking:HF_typeHF_8  0.220          0.536          0.406
## sexM:HF_typeHF_2          0.000          0.001          0.000
## sexM:HF_typeHF_3          0.000          0.001          0.000
## sexM:HF_typeHF_5          0.000          0.001          0.000
## sexM:HF_typeHF_7          0.000          0.001          0.000
## sexM:HF_typeHF_8          0.000          0.001          0.000
##                          actvttyw:HF_HF_8 sM:HF_HF_2 sM:HF_HF_3 sM:HF_HF_5
## day2
## activitystanding
## activitywalking
## sexM
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5

```

```

## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## sexM:HF_typeHF_2          0.001
## sexM:HF_typeHF_3          0.001          0.502
## sexM:HF_typeHF_5          0.001          0.522          0.572
## sexM:HF_typeHF_7          0.001          0.486          0.532          0.553
## sexM:HF_typeHF_8          0.001          0.484          0.530          0.551
##                          sM:HF_HF_7
## day2
## activitystanding
## activitywalking
## sexM
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## sexM:HF_typeHF_2
## sexM:HF_typeHF_3
## sexM:HF_typeHF_5
## sexM:HF_typeHF_7
## sexM:HF_typeHF_8          0.513
##
## Standardized Within-Group Residuals:
##           Min           Q1           Med           Q3           Max
## -5.8180449 -0.4262270  0.1072506  0.5857570  3.2774555
##
## Number of Observations: 2638
## Number of Groups:
##           person HF_type %in% person
##           24           144

```

Section 6: Discussion